

The Energy Spectrometer System

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J.C. Bergstrom

1. INTRODUCTION

An Energy Spectrometer System (ESS) has been installed following the Energy Compression System (ECS). (For historical reasons, this system is also known as the “tune-up line”.) Its function is two-fold: to determine the beam centroid energy, and to determine the beam energy spread. As will be shown, the latter is directly related to the uncertainty in the measurement of the centroid energy. A measurement of the beam energy spread is crucial for fine-tuning the amplitude and phase of the RF accelerating section in the ECS in order to optimize the energy compression. Typically, the ECS reduces the energy spread of the beam from 1-2% as it emerges from the Linac, down to 0.1-0.2% as it enters the LTB transfer line. The beam energy determination is limited by the energy spread within the beam, i.e., 0.1-0.2%. Uncertainties in the calibration of the ESS dipole magnets raise the overall resolution to roughly $\pm 0.3\%$, or about ± 0.7 MeV for a beam energy of 230 MeV.

2. A COMMENT ON BEAM ENERGY

An electron, deflected in a magnetic field B with an orbit radius R , has a momentum p given by the expression

$$pc = ecBR \quad (1)$$

where c is the speed of light, e is the electron charge, and motion is confined to a plane perpendicular to the field. Expressed in useful units, the above becomes

$$pc \text{ (GeV)} = 0.2998 B(T) R(m). \quad (2)$$

Deflection in a magnetic field thus provides a direct measure of the particle momentum. The total particle energy is deduced from the relation

$$E_{\text{total}} = (p^2 c^2 + m^2 c^4)^{1/2} \quad (3)$$

where m is the rest mass ($mc^2 = 0.511 \text{ MeV}$). Since mc^2/pc is small for energies in the few hundred MeV range, we may expand the square root in Eq. (3) and obtain

$$E_{\text{total}} \approx pc + mc^2 \left(\frac{mc^2}{2pc} \right). \quad (4)$$

The second term is a correction at the level of 0.0002% and therefore completely negligible in our energy range (200 MeV). Thus, to a very good approximation, the spectrometer provides a direct measure of the *total* particle energy. (This is definitely not true at lower energies where the second term in Eq. (4) is no longer negligible.)

3. ENERGY RESOLUTION – GENERAL REMARKS

The purpose of our instrumentation (“energy spectrometer”) is to define the beam energy to within a certain degree of uncertainty that we will term the “energy determination resolution”. A precise determination of the beam energy is

hindered by two factors: the energy spread within the beam, and the intrinsic optical characteristics of the spectrometer which limit the energy determination even in the limit of a vanishing energy spread in the beam. Accordingly, we define the following:

γ : the beam energy spread (percent)

Γ : the uncertainty in the beam energy determination (percent)

Note that in accordance with the standard convention, the full-width of these quantities is 2γ and 2Γ , respectively.

We now introduce the so-called “beam matrix” σ , which in the (x, θ) plane is written

$$\sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}. \quad (5)$$

where, from symmetry, $\sigma_{21} = \sigma_{12}$. The diagonal elements describe the maximum extent of the beam in this 2-dimensional phase space:

$$x_{\max} = \sqrt{\sigma_{11}} \quad (6)$$

$$\theta_{\max} = \sqrt{\sigma_{22}}$$

while the off-diagonal elements describe the “tilt” of the phase-space ellipse.

The first-order optical characteristics of the energy spectrometer are described by the conventional transfer matrix R . Neglecting dispersion for the moment, the matrix in our (x, θ) phase space is represented by

$$R = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix}. \quad (7)$$

Although R is not a symmetric matrix, it does satisfy the requirement $\det(R) = 1$ where “det” means “determinant”. (Liouville’s theorem.)

For a given transfer matrix R , the σ -matrix at a point “b” is given by the σ -matrix at a point “a” according to

$$\sigma(b) = R\sigma(a)R^T. \quad (8)$$

where R^T is the transpose of R . From this we deduce the square of the beam size at b:

$$\sigma_{11}(b) = R_{11}^2\sigma_{11}(a) + R_{12}^2\sigma_{22}(a) + 2R_{11}R_{12}\sigma_{21}(a). \quad (9)$$

Let us assume that the source point “a” is located at a beam-defining collimator, such as a pair of slits. Then the phase-space ellipse of the emerging beam will be essentially upright, so $\sigma_{21}(a) \approx 0$. Let us now also allow for some energy dispersion at point “b”, denoted by R_{16} with units of cm/%. Then Eq. (9) takes the form

$$\sigma_{11}(b) = R_{11}^2\sigma_{11}(a) + R_{12}^2\sigma_{22}(a) + R_{16}^2\gamma^2. \quad (10)$$

At location “b” a measurement of the beam profile returns a Gaussian-like function whose standard deviation is roughly given by $\sqrt{\sigma_{11}(b)}$. The beam energy is determined by varying the spectrometer dipole magnets to sweep this Gaussian image across a detector, or across a fixed slit. Let us suppose that a variation of the dipoles by $\pm\Gamma$ is sufficient to discern a variation in intensity across the detector (or slit). Given that the dispersion at the detector (i.e., point “b”) is R_{16} , we have the approximate resolution criterion

$$R_{16}^2 \Gamma^2 = R_{11}^2 \sigma_{11}(a) + R_{12}^2 \sigma_{22}(a) + R_{16}^2 \gamma^2. \quad (11)$$

This is the defining expression for the uncertainty Γ of the beam centroid energy.

The right side of Eq. (11) divides into an achromatic part (γ -independent) and a chromatic part, which we denote as A and B respectively:

$$R_{16}^2 \Gamma^2 = A + B \quad (12)$$

$$A = R_{11}^2 \sigma_{11}(a) + R_{12}^2 \sigma_{22}(a)$$

$$B = R_{16}^2 \gamma^2.$$

Let us consider two extreme scenarios, $A \gg B$ and $B \gg A$ respectively. The former case applies when the energy spread γ is very small and the achromatic focus at “b” is not particularly sharp. The latter case applies when γ is moderate (0.1 – 1.0%) and the achromatic focus is very sharp. The beam energy uncertainty in each case is

$$\Gamma \approx \frac{\sqrt{A}}{R_{16}} \quad \text{if } A \gg B. \quad (13)$$

$$\Gamma \approx \gamma \quad \text{if } B \gg A. \quad (14)$$

Since the energy spread of the beam emerging from the Energy Compression System (ECS) is typically $\gamma = 0.1 - 0.2\%$, the energy spectrometer has been designed to satisfy Eq. (14) instead of Eq. (13). To satisfy Eq. (13) (for $\Gamma \approx 0.1 - 0.2\%$) would require a very large dispersion R_{16} .

The beam energy uncertainty is

$$\Gamma = 0.1 - 0.2\% \quad (15)$$

as constrained by the beam energy spread γ . Of course, the above discussions have assumed that the absolute magnetic calibrations of the system dipole magnets are somewhat better than 0.1%. If not, they will introduce a systematic error into the energy determinations.

4. THE ENERGY SPECTROMETER SYSTEM (ESS)

The ESS is a stripped-down remnant of the now-decommissioned EROS injection beam-line. The physical layout is shown in Fig. 1. It consists of two quadrupole magnets labeled Q_1 and Q_2 , and two 45° dipole magnets. The dipoles have parallel pole edges, and a bend radius $R = 0.800$ m. The three remaining quadrupoles in Fig. 1 are inactive during the ESS operation (but are active during Booster injection).

The remaining components of the ESS are the x-slits, the transition radiation monitor (TRM), and a beam dump. The slits are a part of the original EROS injection line, used to measure the energy spread γ of the injected beam. The TRM has been added in order to get a more quantitative measure of γ (and, by association, Γ).

There are only two focussing “knobs”, Q_1 and Q_2 , and both are wired to focus in the horizontal plane. Quadrupole Q_2 serves two purposes: it compensates the vertical focussing generated by the 22.5° pole-face rotations of the dipole magnets, and it controls the dispersion R_{16} at the detector (x-slit and TRM). The empirical relation between R_{16} and Q_2 is

$$Q_2 = \left(\frac{a - R_{16}}{b} \right) \frac{E}{300} . \quad (16)$$

where Q_2 = pole-tip field in kG, R_{16} = cm/%, E = beam energy in MeV, $a = 3.612$ and $b = 0.644$.

For a given setting of Q_2 (ie., for a given dispersion R_{16}), quadrupole Q_1 is adjusted to minimize the A-term of Eq. (12), where

$$A = R_{11}^2 \sigma_{11}(a) + R_{12}^2 \sigma_{22}(a) . \quad (17)$$

This is equivalent to minimizing the horizontal beam size at the x-slit since, for a monochromatic beam, $x_{\max} = \sqrt{A}$ [recall Eq. (6) and Eq. (10)]. The monochromatic beam size at the slit should be in the sub-millimeter range when Q_1 is optimized.

Although Q_2 *in principle* can control the dispersion over a wide range (ie., $R_{16} = 1.9 - 3.2$ cm/%), *in practice* it is not possible to get a good focus using Q_1 for all values of R_{16} . In particular, the system becomes unstable at $R_{16} = 2.7$ cm/%, in the sense that it is not possible to generate a tight focus for moderate values of Q_1 .

The most favorable values of R_{16} fall within the range

$$R_{16} = 2.0 - 2.5 \text{ cm/\%} . \quad (18)$$

where the optimum Q_1 strengths span the range

$$Q_1 = (0.87 - 1.43) \frac{E}{300} \text{ kG} . \quad (19)$$

The (monochromatic) x-focus improves by about a factor of 4 in passing from the low to high end of R_{16} of Eq. (18), while the corresponding y-focus deteriorates by a factor of 3.

The optimum value for Q_1 will likely be determined empirically by minimizing the actual beam width at the x-slit, so we can only assume that the condition implied by Eq. (14) is indeed satisfied. Then the full beam width, as defined for example by the full aperture of the slit, is given by

$$\text{width} = 2\gamma R_{16} . \quad (20)$$

$$= 2\Gamma R_{16} .$$

which determines Γ . As noted, one should expect

$$\Gamma \approx 0.1 - 0.2\%. \quad (21)$$

5. OPERATIONAL CONSIDERATIONS

- (a) The operator is confronted with two measurements: to determine the beam centroid energy, and to determine the energy spread. The beam energy is determined by changing the dipole strengths in order to sweep the beam across the slit when it is set for a *narrow* aperture. A wider aperture is used to estimate γ . (In principle the quadrupole Q_2 should track with the dipoles, but in practice it is not necessary.)
- (b) Based on previous experience, the absolute energy of the beam should be based on the *mean* field of the two dipole magnets.
- (c) A TRANSPORT file is appended (Appendix A) to illustrate the optical characteristics, assuming a truly monochromatic beam. The beam energy is taken as 300 MeV, and explains the appearance of “300” in Eq. (16) and Eq. (19).

6. ABSOLUTE ENERGY CALIBRATION

As previously noted, this depends on the reliability of the dipole field calibrations. Until such a time as the dipoles are re-calibrated, we fall back on previous experience with EROS. Pion photoproduction at threshold provides an absolute energy marker, and cross-checks indicated that the dipole calibrations were reliable to within plus or minus a few hundred keV, i.e., about $\pm 0.15\%$.

With everything considered, the ESS should be accurate to roughly $\pm 0.3\%$. This translates to an uncertainty in the beam energy of about ± 0.7 MeV for an injection energy to the Booster of 230 MeV. While this precision presents no problems for the Booster, it is marginal for setting the fields of certain magnets in the LTB transfer line.

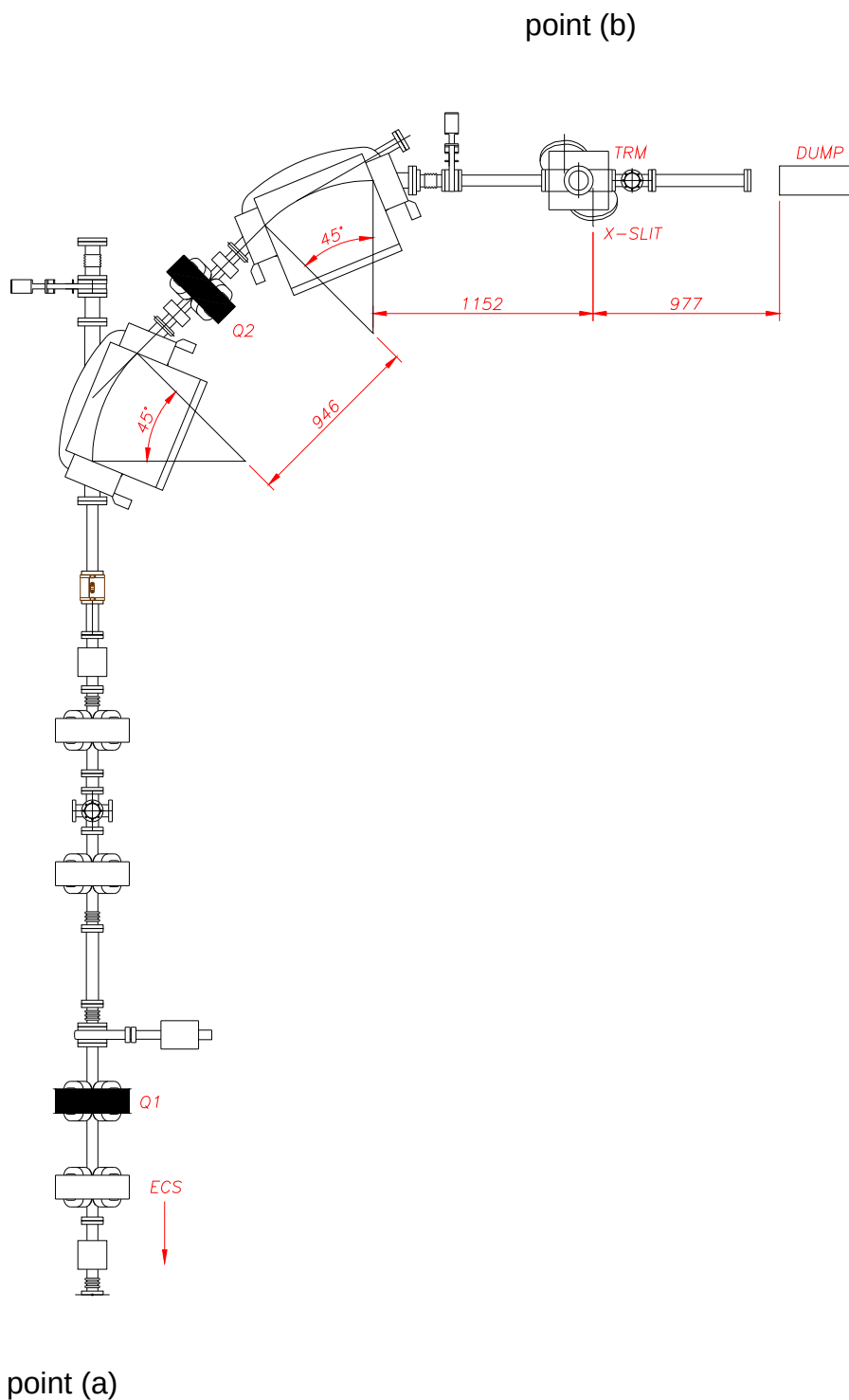


Figure 1: The Energy Spectrometer System (ESS)

Appendix A

ENERGY SPECTROMETER - TUNEUP LINE

0
15. 11. 'mev' 0.001;
1. 0.10 0.288 0.10 0.288 0. 0.000 300. 'bm';
13. 48. 'deg';
3.0 1.60 'dr0';
(-----)
(ECS)
(-----)
16. 5. 2.0 'gap';
16. 7. 0.357 'k1';
16. 2. 0.000 'k0';
2.0 0.0 'rot';
4.0 0.98559 38.940 0.0 'm1';
2.0 38.94 'rot';
3.0 0.831 ;
20. 180.;
2.0 38.94 'rot';
4.0 1.02800 77.88 0.0 'm2';
2.0 38.94 'rot';
20. -180.;
3.0 0.831 ;
2.0 38.94 'rot';
4.0 0.98559 38.94 0.0 'm3';
2.0 0.0 'rot';
3.0 0.650;
3.0 0.60 'accl';
3.0 0.705;
5.00 0.16 -0.00 3.25 ;
3.0 0.29 ;
5.01 0.16 1.442 3.25 'Q1';
3.0 1.03 ;
5.00 0.16 0.00 3.25 ;
3.0 0.59 ;
5.00 0.16 0.00 3.25 ;
3.0 1.327;
16. 5. 1.75 'gap';
16. 7. 0.560 'k1';
2.0 22.5 'rot';
4.00 0.6283 45.00 0. 'ben1';
2.0 22.5 'rot';
3.0 0.3935 ;
5.01 0.15930 1.717 3.25 'Q2';
3.0 0.3935 ;
2.0 22.5 'rot';
4.00 0.6283 45.00 0. 'ben2';
2.0 22.5 'rot';
3.0 1.151;
13. 4. 'slit';
10. 1. 1. 0.01 .001 'x';
10. -1. 6. 2.50 0.001 'r16';
13. 1.;

3.0 1.05;
13. 4. 'dump';
13. 1.;
SENTINEL
SENTINEL